MultiCoNER II

Our approach: knowledge-aware, noise-robust fine-grained NER

SemEval-2023 Task 2 | CS-685 NLP

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The problem

Fine-grained multilingual NER, scored on exact span and exact type

- Input: a short sentence in one of 12 languages
- Output: one BIO tag per token over 33 fine types
- Metric: entity-level macro-F1
- Reference: organizers' XLM-R baseline is about 0.63

Low context

"Jordan won again"

Fine types

Scientist vs Athlete vs Politician

Missing knowledge

"Dark" is a TV series

Noisy test data

Typos in entities and context

Example

Lionel	Messi	joined	Inter	Miami
B-Athlete	I-Athlete	O	B-SportsGRP	I-SportsGRP

What prior work leaves open

Each design choice answers a weakness in the closest systems

Closest work	What it shows	Weakness	Our response
DAMO U-RaNER	Rich Wikidata retrieval won 9 of 13 tracks	Heavy pipeline, big ensemble	Single encoder, top-k context, small ensemble
PAI	Simple retrieval + concatenation is strong	Retrieval noise goes straight into the encoder	Relevance filter, knowledge dropout, learned gate
GEMNET / dynamic gazetteers	Gated gazetteer features help low context	Exact matching breaks on typos	Fuzzy + dense retrieval, noise augmentation
XLM-R baseline	About 0.63 macro-F1	No outside knowledge, weak on noise	Used as A0, the number to beat

Our approach at a glance

One encoder, filtered retrieval, noise-aware training, light ensemble



Our additions beyond the plain baseline

Add knowledge

Wikipedia + Wikidata retrieval gives context for rare entities

Trust it selectively

Filter and gate so wrong retrieval does not hurt

Survive noise

Train on typo-corrupted text; keep cost low

Worked example: one sentence, end to end

How each stage turns a hard, low-context input into the right tags

1 Input

“dark was released in 2017”

Lowercase, no context. A plain encoder sees a common word.

2 Retrieve

Search on the span “dark”

Fuzzy + dense retrieval returns Wikidata entries with label, description and type.

3 Filter

Re-rank, keep top entries

The 2017 TV series “Dark” stays; unrelated senses fall below the threshold.

4 Fuse + tag

Gate leans on the type hint

Context is thin, so knowledge gets more weight; the CRF decodes the tags.

Output

dark	was	released	in	2017
B-VisualWork	O	O	O	O

Noise check: with the typo “drak was released in 2017”, fuzzy retrieval still reaches “Dark”, and noise training teaches the model to keep the same label.

Illustrative example, not a reported result.

What we are working on

Baseline first, then one component at a time so every gain is measured

Focus	Stages	What we are doing	Ablation
Now	S1-S2	Label maps, BIO repair, language sampling; XLM-R + CRF baseline	A0
Now	S3	Building the Wikipedia + Wikidata index; offline cached retrieval	A1
Next	S4	Relevance filter and context builder	A2
Next	S5-S6	Noise augmentation, gated fusion, coarse head	A3, A4
Later	S7	Seed ensemble, per-language and noisy-vs-clean analysis	A5

Each ablation adds one component: A0 baseline, A1 naive retrieval, A2 filtering, A3 noise training, A4 gating, A5 ensemble.

How the model works

Retrieve, filter, then let the model learn when to trust the knowledge

Retrieve and filter (S3-S4)

- Offline KB: Wikidata label, aliases, description, type; Wikipedia leads in 12 languages
- BM25 + multilingual dense search; fuzzy match so typos still hit
- Keep top 3-5 above a threshold, max 256 context tokens

Gated fusion (S6)

- Sentence + context go through a shared XLM-R-large encoder
- Type-hint embeddings k are mixed with encoder states h: z = g*k + (1-g)*h
- Gate falls back to h when knowledge looks wrong

Noise-robust training (S5)

- Typo, swap and drop noise, adapted per script
- Corrupt entities and context like the noisy test set
- Loss = fine + coarse + no-context + KL(clean || noisy)

Fine head (67 BIO tags, CRF) plus an auxiliary coarse head (13 tags) that regularizes the 33-way decision.